NTU Physics Discovery Camp
Computational Nonlinear Lab: Chaos with Logistic Map

June 2026

> *"You think you know when you learn, are more sure when you can write, even more when you can teach, but certain when you can program."*
>
> *— Alan Perlis*
> *First recipient of the Turing Award*

1. Introduction

Supposed that we are concerned with how the population of a biological species (e.g., fruit flies, ants, etc.) changes every year. How would we go about to model this?

In 1976, the biologist Robert May applied a simple mathematical model called the *logistic map* to model such population dynamics, which unveils surprising results. Today, we will look at these results, and illustrate how simple models can give rise to bewildering complex behavior and chaos.

Intuitively, if we know that there are N_0 populations in year 0, we might model the population N_1 in year 1 as

$$N_1 = AN_0, \tag{1}$$

where A is a parameter that depends on the conditions of the environment, such as food and water supplies, diseases, or environmental conditions etc. We assume that A remains the same for subsequent generations (i.e., it is constant in year 1, year 2, ..., year n, etc.).

Question 1*: What will happen to the population if (a) $A < 1$ and (b) $A > 1$?*

- (a) $N_\infty \to 0$
- (b) $N_\infty \to \infty$

However, this model is too simple; we know that if the population grows too large, there will not be enough food to support the larger population, or perhaps predators are more likely to hunt them for food. Hence, realistically, the population growth should be limited.

This feature can be incorporated into the model by adding another term that is unimportant for small population N, but becomes more important when N increases. One way of doing this is to introduce a term proportional to N^2 as follows:

$$N_{n+1} = AN_n - BN_n^2. \tag{2}$$

If $B \ll A$, then the second term in the above equation will not be important until N gets sufficiently large. The minus sign means that the second term tends to decrease the population.

But of course it does not make sense for population N_{n+1} to be a negative number. As such, there must be some constraint on A, B, and the maximum population number $N_{\max}$, which if violated, will give us the unphysical result of $N_{n+1} < 0$.

***Question 2**: Derive that in order for $N_{n+1} \geq 0$, the constraint for the maximum population number should be $N_{\max} = \frac{A}{B}$.*

$$\begin{aligned} N_{n+1} = AN_n - BN_n^2 &\geq 0 \\ AN_n &\geq BN_n^2 \\ A &\geq BN_n \\ \frac{A}{B} &\geq N_n. \end{aligned}$$

Therefore, for $N_{n+1} \geq 0$, the upper bound of N_n is $\frac{A}{B}$ or $N_{\max} = \frac{A}{B}$.

Next, let us normalize the equation $N_{n+1} = AN_n - BN_n^2$ by $N_{\max}$ and denote $x_n = \frac{N_n}{N_{\max}}$. We arrive, after some algebraic manipulation, the logistic map equation:

$$x_{n+1} = Ax_n(1 - x_n) \tag{3}$$

with $x_n \in [0, 1]$. This equation plays a very important role in the development of the theory of chaos.

Note that x_n is the population (as a fraction of $N_{\max}$) in the n-th year. This is useful as now we only have to specify a single parameter A, and we do not have to specify a maximum population $N_{\max}$, which can be different for different species.

***Question 3**: Show the derivation of $x_{n+1} = Ax_n(1 - x_n)$.*

Normalizing $N_{n+1} = AN_n - BN_n^2$ with $N_{\max}$, and letting $x_n = N_n / N_{\max}$,

$$\begin{aligned} \frac{N_{n+1}}{N_{\max}} &= A\frac{N_n}{N_{\max}} - B\frac{N_n^2}{N_{\max}} \\ x_{n+1} &= Ax_n - Bx_nN_n \\ &= Ax_n\left(1 - \frac{B}{A}N_n\right) \\ &= Ax_n\left(1 - \frac{1}{N_{\max}}N_n\right) \\ &= Ax_n(1 - x_n). \end{aligned}$$

2. Fixed Points and Stability

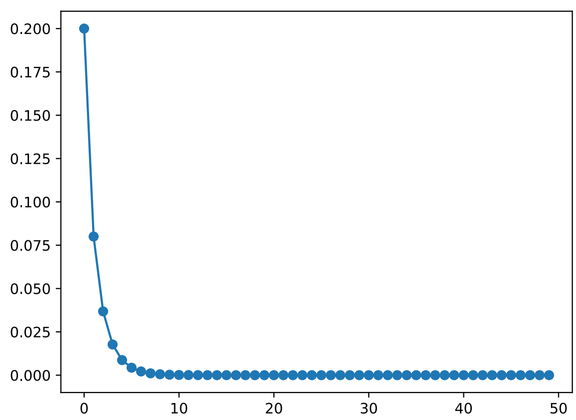
Next, we shall adjust the parameter A and observe how this affect the population dynamics. We shall restrict A to the range $0 \le A \le 4$ so that Eq. (3) will map the interval $0 \le x_n \le 1$ to itself.

***Computation 1**: Start from year 0 or x_0 with an arbitrary population, and iterate $x_{n+1} = Ax_n(1 - x_n)$ with $A = 0.5$ for many years. Repeat the same for $A = 2$. Observe and comment on what happens as the number of years n increase for the two different values of A.*

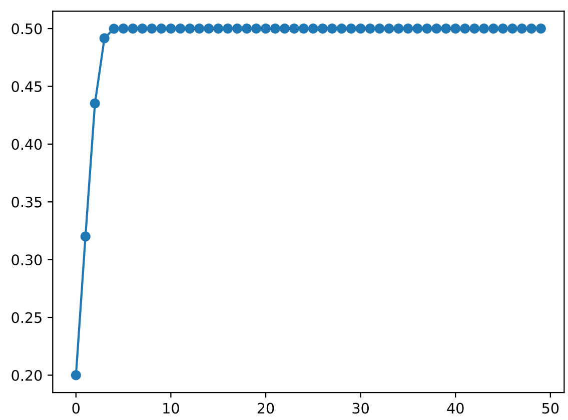
Computation 1 Python

```python
import numpy as np
import matplotlib.pyplot as plt

x = np.zeros(50)  # Initialize x_0 to x_49. This also set them to be 0
x[0] = 0.2        # Set x_0 = 0.2

A = 0.5
for n in range(49):  # iterate n over [0, 49)
    x[n+1] = A * x[n] * (1 - x[n])  # The logistic map

plt.plot(x, 'o-')  # Plot with the symbol 'o-'
```

$A = 0.5$



$A = 2$



Instructor's note

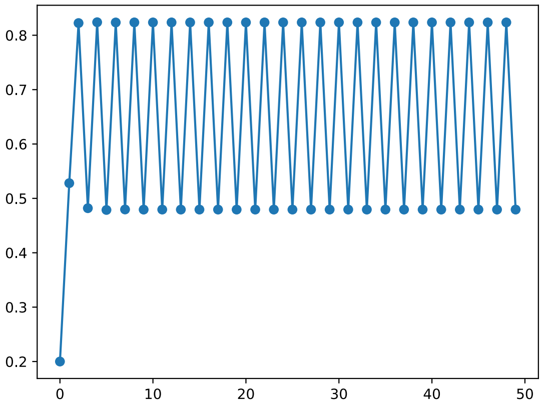
Explain the concept of fixed point and its stability for the respective cases of $A = 0.5$ and $A = 2$.

As expected, if $A < 1$, the population will eventually dies out.

On the other hand, for $A = 2$, the population will reach a maximum number and cannot grow to infinity.

***Computation 2**: Repeat the same for $A = 3.3$. Observe and comment on what happens as the number of years n increase.*

$$A = 3.3$$



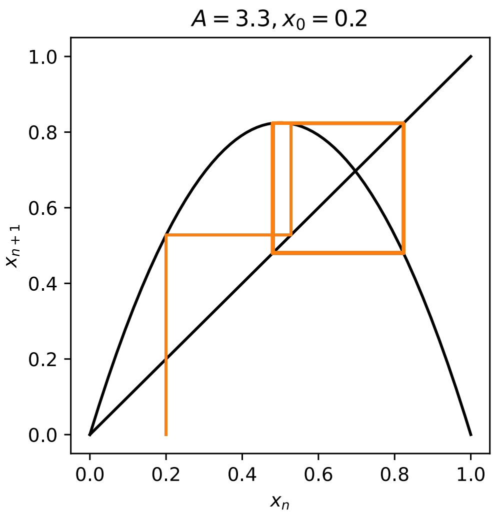
Instructor's note

We can check the values that x_n oscillates between with

```python
x[-1], x[-2]
```

```
(0.47942701982423314, 0.8236032832060693)
```

Explain the second-iterate map and period-2 cycle. Demonstrate with a cobweb diagram.



3. Bifurcation Diagram

An *attractor* of a dynamical system is the set of points at which the system eventually settles into. Let us next examine the *bifurcation diagram* of the logistic map, which is a diagram of the system's attractor as a function of A.

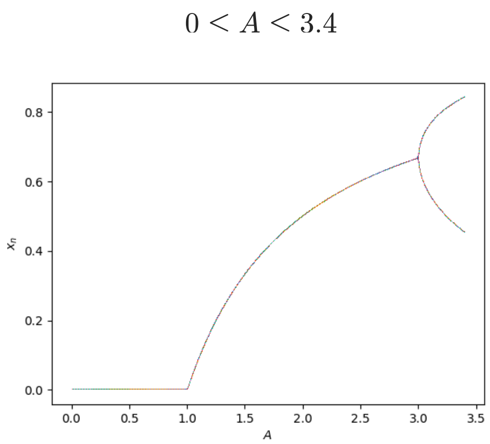
Earlier, we observed that for different values of A, x_n reached different fixed points or cycles. The bifurcation diagram is simply a plot of these fixed points and cycles against A.

> 🔥 **Tip**
>
> To generate the bifurcation diagram,
>
> 1. First, loop through the range of values of A.
> 2. Then, for each value of A in the loop, iterate the logistic map for many years, say from x_0 to x_{599}, to allow the system to settle down to its eventual behavior.
> 3. Next, plot many values of x_n against the corresponding A for values of n after it settles down to its eventual behavior, say $x_{300}, \ldots, x_{599}$.
> 4. The plot function `plt.plot()` can take an array of x values and y values to plot. E.g., try running `plt.plot([0.5] * 5, [1, 2, 3, 4, 5], '.')`, which plot the points (0.5, 1), (0.5, 2), (0.5, 3), (0.5, 4), and (0.5, 5).

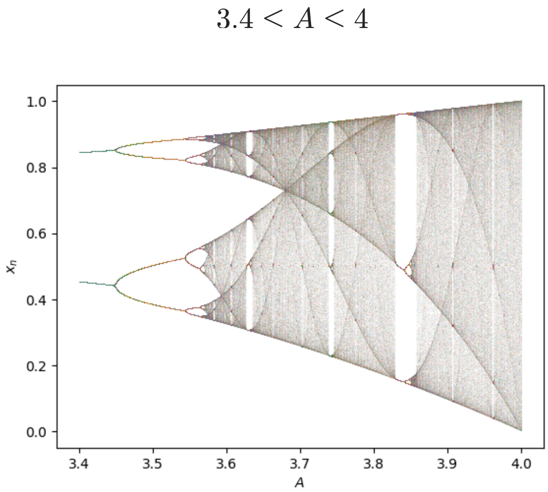
Computation 3: *Plot the bifurcation diagram for the range $0 \leq A \leq 3.4$.*

Computation 3 Python

```python
for A in np.linspace(0, 3.4, 3000):  # loop through many values of A
    x = np.zeros(600)
    x[0] = 0.2

    for n in range(599):
        x[n+1] = A * x[n] * (1 - x[n])

    plt.plot([A] * 300, x[300:], ',')  # Plot (A, x[300]), (A, x[301]), etc.

plt.xlabel("$A$")
plt.ylabel("$x_n$")
```



Computation 4: *Repeat Computation 3 for the range* $3.4 \leq A \leq 4$.

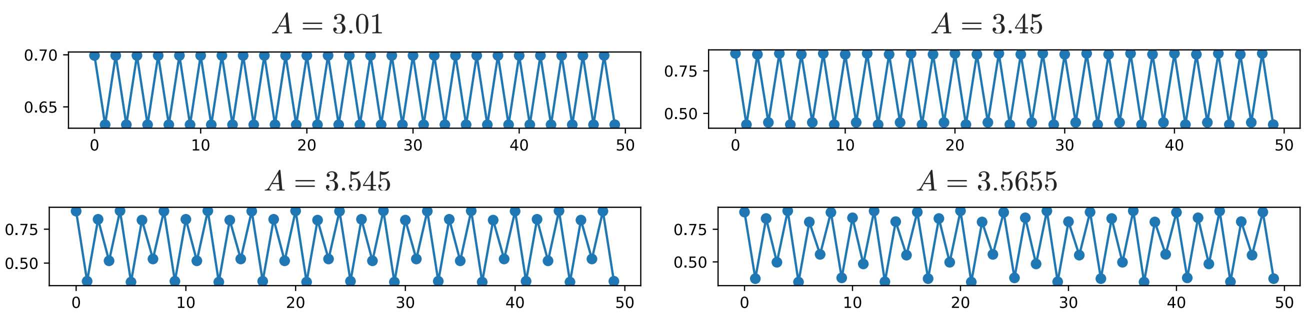


Observe the presence of a period-doubling in the plotted bifurcation diagram. The bifurcation can be examined in greater detail through the following computation.

Computation 5: *Compute the steady-state dynamics for* $A_1 = 3.01$, $A_2 = 3.45$, $A_3 = 3.545$ *and* $A_4 = 3.5655$ *like in Computations 1 and 2. Infer the period of the bifurcation in each case.*

Computation 5 Python

```python
for A in [3.01, 3.45, 3.545, 3.5655]:

    x = np.zeros(600)
    x[0] = 0.2

    for n in range(599):
        x[n+1] = A * x[n] * (1 - x[n])

    # We create a new figure for each value of A
    plt.figure(figsize=(6, 1))
    plt.plot(x[-50:], '.-')  # We plot only the last 50 values
    print(x[-20:])           # We can also print out the last 20 values
```



Instructor's note

```
[0.69937714 0.63284875 0.69937714 0.63284876 0.69937713 0.63284876
 0.69937713 0.63284877 0.69937713 0.63284877 0.69937712 0.63284877
 0.69937712 0.63284878 0.69937712 0.63284878 0.69937712 0.63284878
 0.69937711 0.63284879]
[0.84732753 0.44630437 0.85255289 0.43368729 0.84732905 0.44630072
 0.85255154 0.43369058 0.84733056 0.44629712 0.8525502  0.43369383
 0.84733205 0.44629355 0.85254888 0.43369704 0.84733352 0.44629003
 0.85254758 0.43370021]
[0.8168705  0.5303075  0.88299376 0.36625444 0.82283749 0.51677577
 0.88525234 0.36010334 0.81687054 0.5303074  0.88299378 0.36625438
 0.82283743 0.51677589 0.88525233 0.36010338 0.81687058 0.53030731
 0.8829938  0.36625433]
[0.80863719 0.55173647 0.88183136 0.37154226 0.83253929 0.49709353
 0.89134488 0.34531577 0.80606253 0.55737939 0.87963597 0.37750277
 0.83787264 0.48434494 0.89050116 0.34766777 0.80863719 0.55173647
 0.88183136 0.37154226]
```

We also print out the last 20 values as it is difficult to determine the period of the bifurcation from the plots alone. We can infer the period of the bifurcation by counting the number of cycles before a value is repeated again (note that the values might not be exact due to numerical errors).

We should observe period doubling where the period goes from $2 \to 4 \to 8 \to 16$.

Demonstrate the fractal property of the bifurcation diagram.

Question 4: *Evaluate ratio of the difference between how fast period-doubling occur:*

$$\delta_3 = \frac{A_3 - A_2}{A_4 - A_3}. \tag{4}$$

We simply substitute the values for A_2, A_3, and A_4 in:

$$\delta_3 = \frac{A_3 - A_2}{A_4 - A_3} = \frac{3.545 - 3.45}{3.5655 - 3.545} = \frac{0.095}{0.0205} \approx 4.634$$

It is important to note that the successive bifurcations come faster and faster. Ultimately, the A_m converge to a limiting value of $A_\infty \approx 3.5699...$. The convergence is essentially geometric: in the limit of large m, the distance between transitions shrinks by a constant factor

$$\delta = \lim_{m\to\infty} \frac{A_m - A_{m-1}}{A_{m+1} - A_m} = 4.669... \tag{5}$$

Interestingly, this convergence factor δ is universal, and is called the Feigenbaum constant.

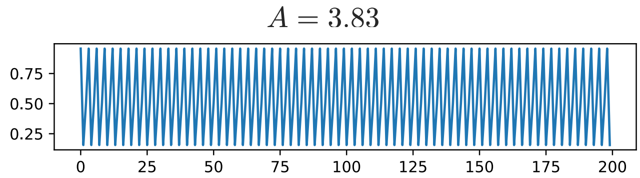
4. Chaotic Dynamics

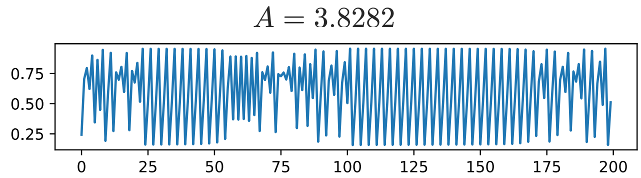
At $A_\infty \approx 3.5699...$, the dynamics of the logistic map is chaotic. The manner in which chaos is reached as A approaches A_∞ from below is known as the *period-doubling route to chaos*. Let us go beyond A_∞ and examine the dynamics of logistic map there.

Computation 6: *Compute the steady state dynamics of the logistic map for $A = 3.83$ and $A = 3.8282$ like in Computations 1 and 2.*

Computation 6 Python

```python
for A in [3.83, 3.8282]:
    x = np.zeros(600)
    x[0] = 0.2

    for n in range(599):
        x[n+1] = A * x[n] * (1 - x[n])

    plt.figure(figsize=(6, 1))
    plt.plot(x[-200:], '-')  # We only plot the last 200 points
```



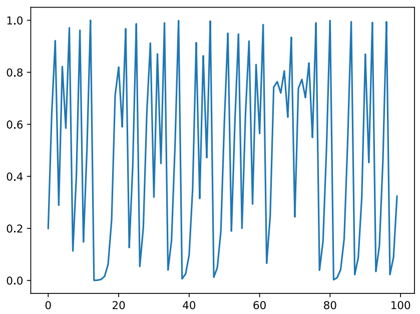


As A_∞ is approached from above, we observe the *intermittency route to chaos*.

Computation 7: *Repeat the same for $A = 4$ for a particular x_0 and observe the aperiodic dynamics.*

Computation 7 Python

```python
x = np.zeros(600)
x[0] = 0.2

A = 4
for n in range(599):
    x[n+1] = A * x[n] * (1 - x[n])

plt.plot(x[:100], '-')  # We plot just the last 100 values
```

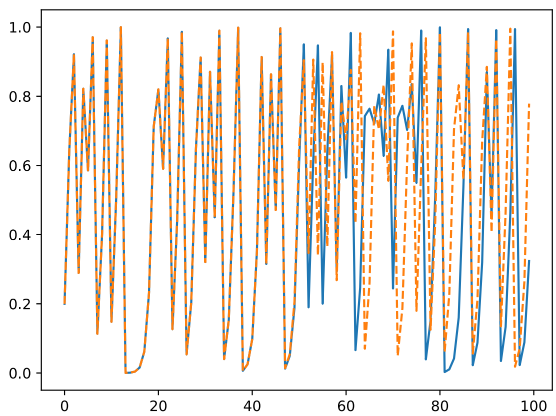


***Computation 8**: Repeat Computation 7 but with a slightly perturbed initial condition $x_0 + \epsilon$ with ϵ a very small number, and compare the dynamics. Notice the fast divergence of the results even though ϵ is small.*

By putting the case of $x_0 + \epsilon$ in the same code block as the previous computation, we can plot both cases in the same plot to see their differences.

Computation 8 Python

```python
# Case of x_0 = 0.2
x = np.zeros(600)
x[0] = 0.2

A = 4
for n in range(599):
    x[n+1] = A * x[n] * (1 - x[n])

plt.plot(x[:100], '-')

# Case of x_0 = 0.2 + 10^-16
x = np.zeros(600)
x[0] = 0.2 + 1e-16

A = 4
for n in range(599):
    x[n+1] = A * x[n] * (1 - x[n])

plt.plot(x[:100], '--')
```



We observe that even though the initial condition x_0 changes by an extremely small amount of just 10^{-16}, its effects can become large eventually.

The trademark of chaos is its sensitive dependence on initial conditions, also more widely known as the “butterfly effect”.